GIS Model for Assessing the Flood Susceptibility of Kazakhstan's Cities: Terrain, Drainage and Land Use

Diploma project · Pre-defense review of research, analysis, requirements, design and planning

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Supervisor: Tulebayev Yersultan, Senior Lecturer, School of Software Engineering

Astana IT University · October 2026

Problem, goal and expected results

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TASK

Problem statement

Spring snowmelt floods hit Kazakhstan's cities almost every year. In April 2024 nearly 100,000 people were evacuated across six regions.

Residents and planners have no open, city-level map that shows which districts are susceptible and, crucially, why.

Global flood maps are coarse and black-box; local studies stay in papers and are not usable by non-experts.

Goal

Build an interpretable GIS model that maps flood susceptibility of districts in several flood-prone cities of Kazakhstan, explains each factor's contribution and lets users compare cities.

Expected results

- Susceptibility maps (5 classes) for 3–4 cities
- Per-factor explanation for any location
- City-to-city comparison report
- Web prototype with an interactive map
- Validated model: AUC and error analysis

Team, target users and scope

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Rakhymbay Yerkebulan — QA engineer & developer

Data preparation, factor calculation in Python, model testing and validation against satellite flood extents

Tenglesbayev Zhandos — Developer

Model implementation, susceptibility and exposure maps, web map prototype

Tutkabay Assem — Analyst & project manager

Literature review, factor justification and weighting, comparative analysis, schedule

Supervisor: Tulebayev Yersultan Bakhytovich

Target users

- Akimat urban planning departments
- Emergency services (MES) planners
- Residents choosing or insuring housing
- Researchers and students

Scope

In: susceptibility (where floods are likely), 8–10 factors, 3–4 cities (candidates: Astana, Petropavl, Atyrau, Oral), open data, web map.

Out: real-time forecasting and flood warnings (requested in the survey, future work), hydrodynamic simulation, damage in money.

Why it matters: the spring 2024 floods

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99,869

people evacuated, incl. 36,591 children

6

regions with flooded homes

66

settlements cut off from roads

Atyrau, Aktobe, Akmola, Kostanay, East Kazakhstan and North Kazakhstan regions, as of 12 April 2024 (Ministry of Emergency Situations via The Astana Times).

What this means for the project

- Floods are recurrent and city-scale, so planners and residents need district-level information before the season, not only during it.
- Kazakh studies (Yertis basin, Ust-Kamenogorsk) confirm the hazard but publish static maps, without explanations or a usable tool.

Survey: who answered and what they said

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Online bilingual survey (RU/EN), 14 questions, 14–15 Oct 2026. Respondents: 20 students, 2 city residents; 18 from Astana, 4 from Almaty, Shymkent, Turkestan and Karaganda.

22
responses so far

55%
affected by floods personally or in their city

95%
have no official info on whether their district is flood-prone

13
volunteered for a follow-up interview

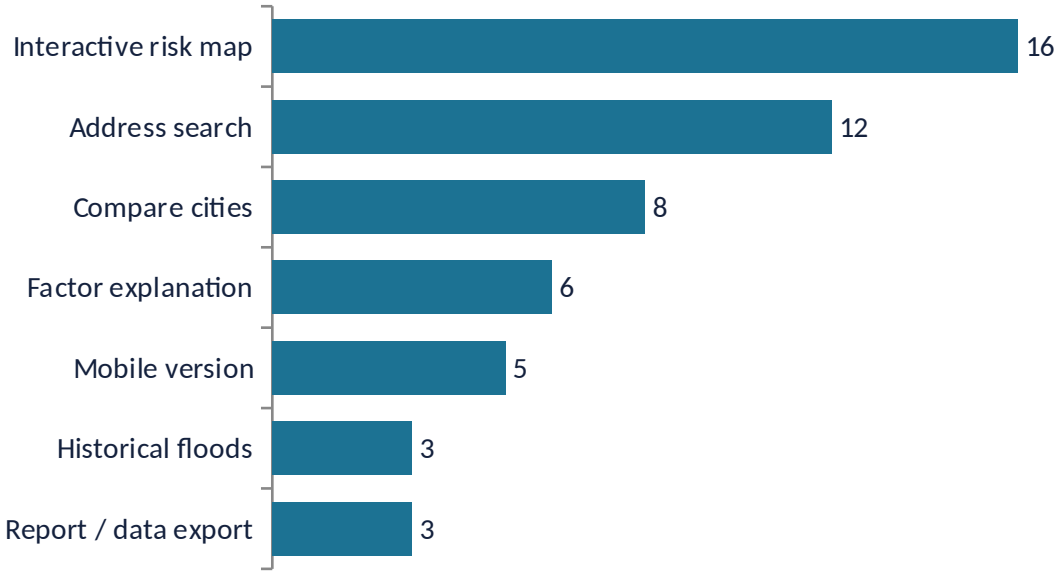
Hypotheses

H1	Most people do not know if their district is flood-prone	**Confirmed**	21 of 22 have no official source; 11 do not know at all
H2	An open district map is useful	**Partly**	Avg 3.82 of 5; 50% rate it 4–5, 41% neutral
H3	Users want to know WHY an area is risky	**Confirmed**	Avg 4.14 of 5; 68% rate it 4–5

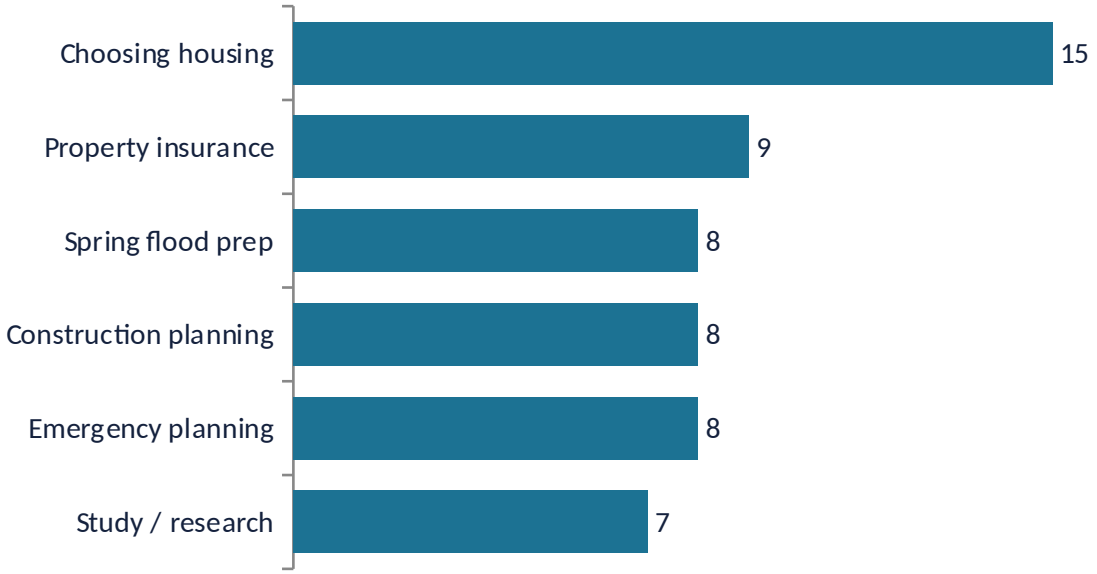
Survey: what users need, and what we changed

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Most important features (top 3, n=22)



What they would use the map for (n=22)



Impact on the project

- Address search raised from Should to Must (FR4); Kazakh/Russian/English UI raised from Could to Should (FR7): RU 77%, KZ 59%, EN 45%.
- Real-time flood warnings were requested in 4 open answers; they are out of scope (forecasting) and listed as future work.
- Limitation: sample is mostly Astana students. Next: interviews with the 13 volunteers and outreach to MES and akimat staff.

Literature review

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#	Authors, year, source	Full title	Method / data	What we use
1	**Saaty, 1990** *Eur. J. Operational Research*	How to make a decision: The analytic hierarchy process	AHP theory	Expert factor weights (baseline)
2	**Rahmati et al., 2016** *Geocarto International*	Flood susceptibility mapping using frequency ratio and weights-of-evidence models in the Golastan Province, Iran	FR, WoE; Iran	Statistical baseline model
3	**Tehrany et al., 2014** *Journal of Hydrology*	Flood susceptibility mapping using a novel ensemble weights-of-evidence and support vector machine models in GIS	WoE + SVM; Malaysia	Ensemble, factor ranking
4	**Khosravi et al., 2018** *Sci. of the Total Environment*	A comparative assessment of decision trees algorithms for flash flood susceptibility modeling at Haraz watershed, northern Iran	Tree-based ML; Iran	Tree models for main model
5	**Pham et al., 2021** *Journal of Hydrology*	Flood risk assessment using hybrid artificial intelligence models integrated with multi-criteria decision analysis in Quang Nam Province, Vietnam	Hybrid AI + MCDA; Vietnam	Combining MCDA with ML
6	**Lundberg & Lee, 2017** *NeurIPS 2017*	A unified approach to interpreting model predictions	SHAP values	Per-factor explanations
7	**Bentivoglio et al., 2022** *Hydrol. & Earth System Sci.*	Deep learning methods for flood mapping: a review of existing applications and future research directions	Review	Gap: interpretability
8	**Twele et al., 2016** *Int. J. Remote Sensing*	Sentinel-1-based flood mapping: a fully automated processing chain	Sentinel-1 SAR	Flood extents for validation
9	**Tellman et al., 2021** *Nature*	Satellite imaging reveals increased proportion of population exposed to floods	MODIS, 2000–2018	Motivation: rising exposure
10	**Ward et al., 2015** *Nature Climate Change*	Usefulness and limitations of global flood risk models	Global models	Why a local model is needed
11	**Apshikur et al., 2023** *ISPRS Archives*	The method of multi-criteria analysis for determining the flood-hazardous area and the development of protective structures	GIS MCA; Ust-Kamenogorsk	Kazakhstan context
12	**Makhmudova et al., 2026** *Environments (MDPI)*	Modeling flood-prone areas using statistical and GIS approaches: Insights from the Yertis River basin, Kazakhstan	DEM scenarios; Yertis basin	Flood zones of KZ cities

12 peer-reviewed publications. Also used: The Astana Times (12 Apr 2024), Copernicus GloFAS, WRI Aqueduct Floods. Gap: no open, interpretable, district-level tool for Kazakhstan's cities.

Comparison with existing systems

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Criterion	GloFAS	Aqueduct Floods	ThinkHazard!	Kazhydromet	Our system
Purpose	Forecast	Risk maps	Hazard level	Bulletins	Susceptibility
District-level detail in cities	No	No	No	No	Yes
Explains contributing factors	No	No	Partly	No	Yes
Compares cities	No	Partly	No	No	Yes
Interactive web map	Yes	Yes	Yes	Partly	Yes
Free and open	Yes	Yes	Yes	Yes	Yes
Kazakh / Russian interface	No	No	No	Yes	Planned
Focus on Kazakhstan	No	No	No	Yes	Yes

Ratings are our assessment from public documentation; to be re-checked during the review.

Why a new solution is needed

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Too coarse

Global products work at river-basin or province scale, so a district inside Astana or Petropavl gets one value.

Black box

Users see a risk class but not the reason: terrain, river proximity, soil or land cover.

Forecast ≠ susceptibility

GloFAS and bulletins warn about a specific event; they do not show where floods are structurally likely.

No local tool

Kazakh research results stay in papers; nothing lets a planner query and compare cities.

Our answer: an open, district-level susceptibility map for Kazakhstan’s cities that explains every result.

Technology choices (1/2)

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Susceptibility method

AHP (expert weights)

+ Simple, transparent, no flood inventory needed
− Subjective weights, no accuracy check

Frequency Ratio

+ Statistical, easy to explain per class
− Ignores factor interactions

✓ Random Forest + SHAP

+ High accuracy, SHAP explains every cell
− Needs inventory, more complex

Choice: Random Forest + SHAP as main model; AHP and Frequency Ratio kept as interpretable baselines.

GIS processing stack

✓ QGIS + Python

+ Free, scriptable (GDAL, rasterio, geopandas)
− Manual setup, local compute

ArcGIS Pro

+ Rich toolset, industry standard
− Paid licence, Windows only

Google Earth Engine

+ Petabyte catalogue, fast
− Vendor lock-in, quotas

Choice: QGIS + Python for the pipeline; Earth Engine only to download Sentinel-1 flood extents.

Technology choices (2/2)

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Spatial data storage

✓ PostgreSQL + PostGIS

+ Spatial queries, multi-user, standard
− Needs a server and admin

GeoPackage + COG files

+ Zero setup, portable
− No concurrent queries

Cloud object storage

+ Scales, cheap for rasters
− Cost, internet dependency

Choice: PostGIS for districts and results; Cloud-Optimized GeoTIFF files for factor rasters.

Component communication & web map

✓ FastAPI REST + Leaflet

+ Python end to end, auto OpenAPI docs
− Async learning curve

Flask + OpenLayers

+ Mature, many examples
− More boilerplate, heavier client

Static map tiles only

+ Cheapest hosting
− No queries or explanations

Choice: FastAPI REST API (JSON + XYZ tiles) with a Leaflet front end; testable with Playwright and REST tests.

Functional requirements

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ID	Requirement	Priority	Acceptance criterion
FR1	Show a susceptibility map (5 classes) for a selected city	Must	Map loads in ≤ 3 s for each of the 3–4 cities
FR2	Explain factor contributions for any clicked location	Must	Panel lists all factors with SHAP share summing to 100%
FR3	Compare cities by share of area in each class	Must	Chart matches values computed from rasters (±1%)
FR4	Search by address or coordinates	Must ↑	Top-1 geocoding hit for 9 of 10 test addresses
FR5	Toggle layers: factors, 2024 flood extent	Should	Each layer toggles without page reload
FR6	Export a city report (PDF) and data (GeoTIFF)	Could	Files open in a PDF reader and QGIS
FR7	Interface in Kazakh, Russian and English	Should ↑	All UI strings translated, switch persists
FR8	Re-run the model with new data (admin)	Won't now	Deferred to after the diploma

Prioritised with MoSCoW. ↑ = raised after the user survey (n=22): address search was a top-3 feature for 55%.

Non-functional requirements

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Accuracy
Must

Model AUC ≥ 0.80 on held-out flood points

Interpretability
Must

Every result has a factor explanation

Performance
Should

Map tiles ≤ 1 s, API p95 ≤ 500 ms

Usability
Should

SUS score ≥ 70 with 5+ test users

Reproducibility
Must

Pipeline re-runs from raw data with one command

Compatibility
Should

Works in Chrome, Firefox, Safari; mobile width 375 px

Security
Must

No personal data stored; HTTPS only; secrets out of Git

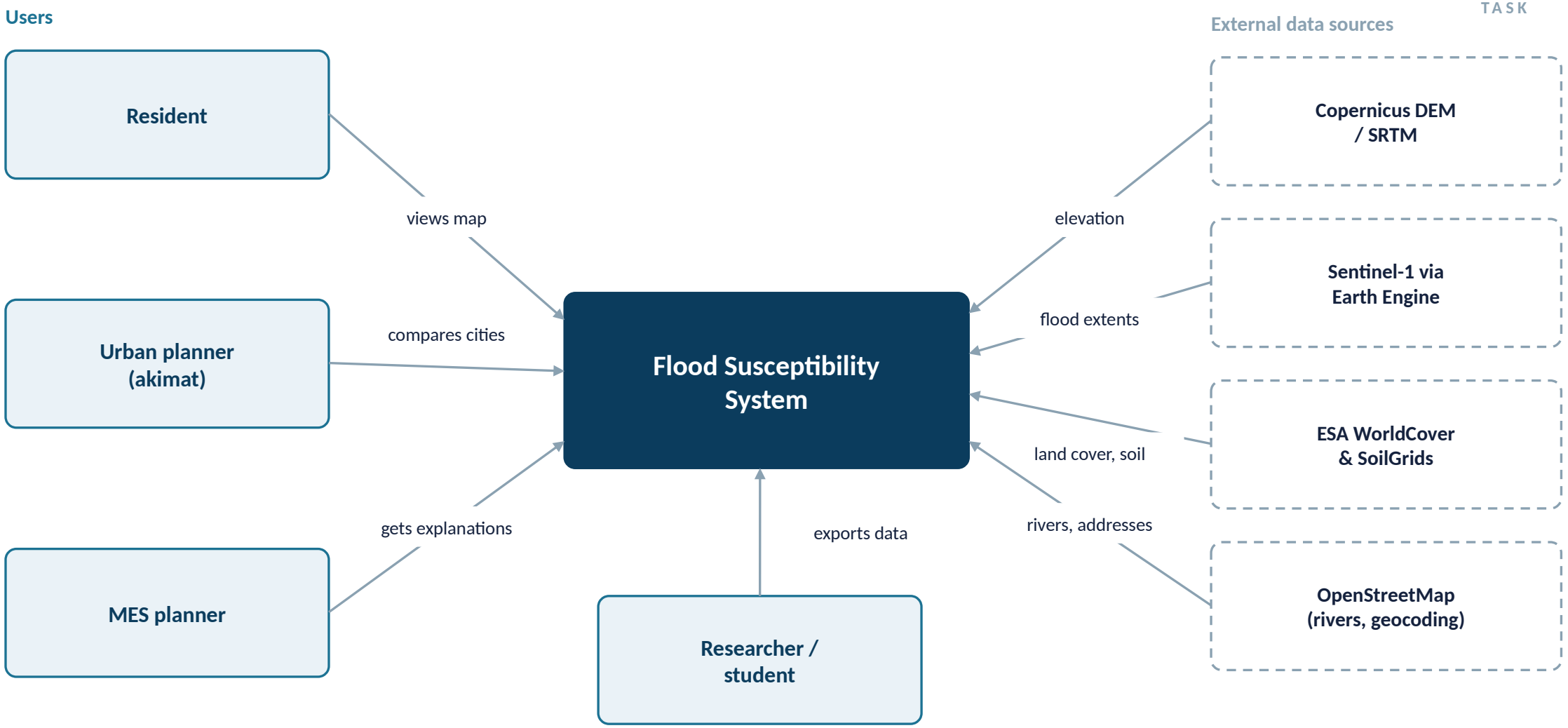
Maintainability
Should

Test coverage ≥ 70% for API and pipeline

System context diagram

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TASK

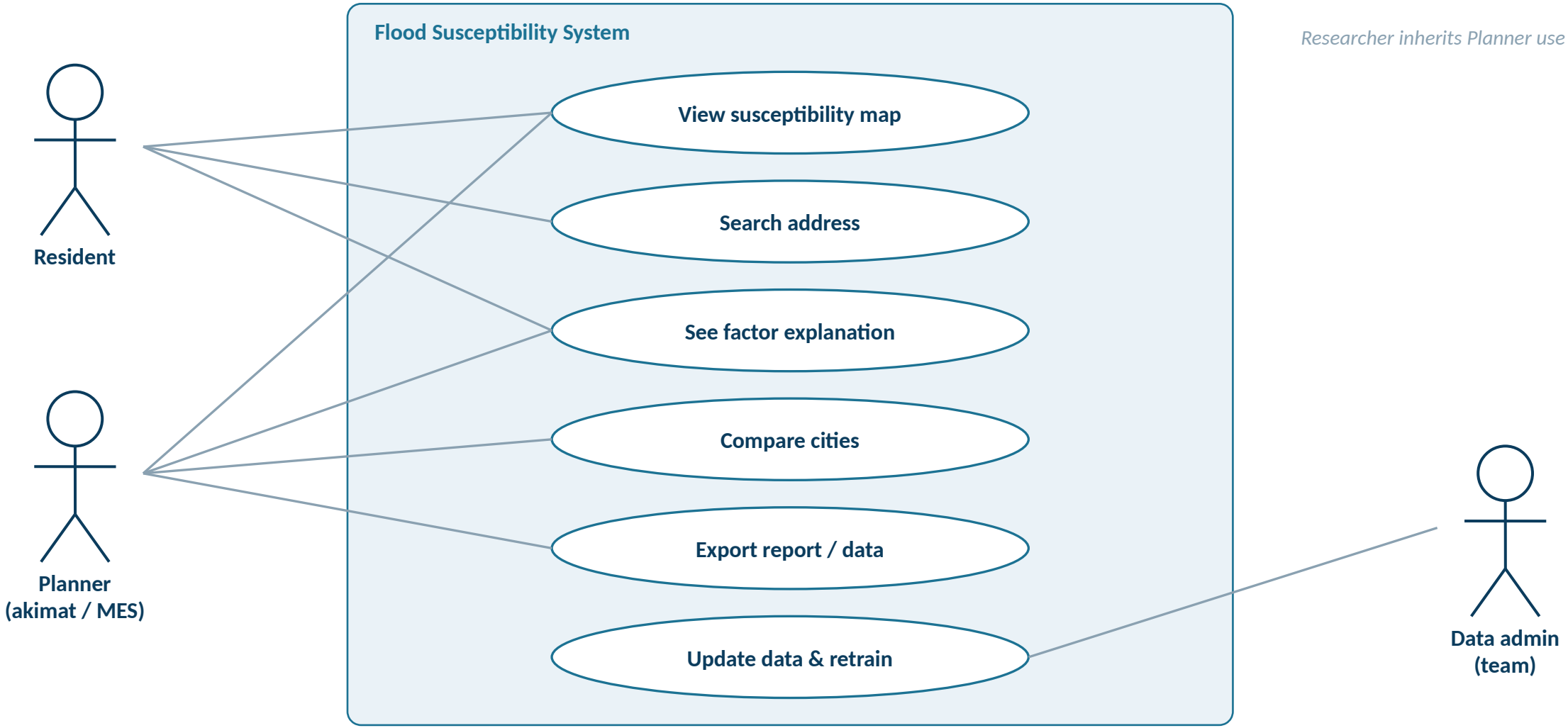


Use case diagram

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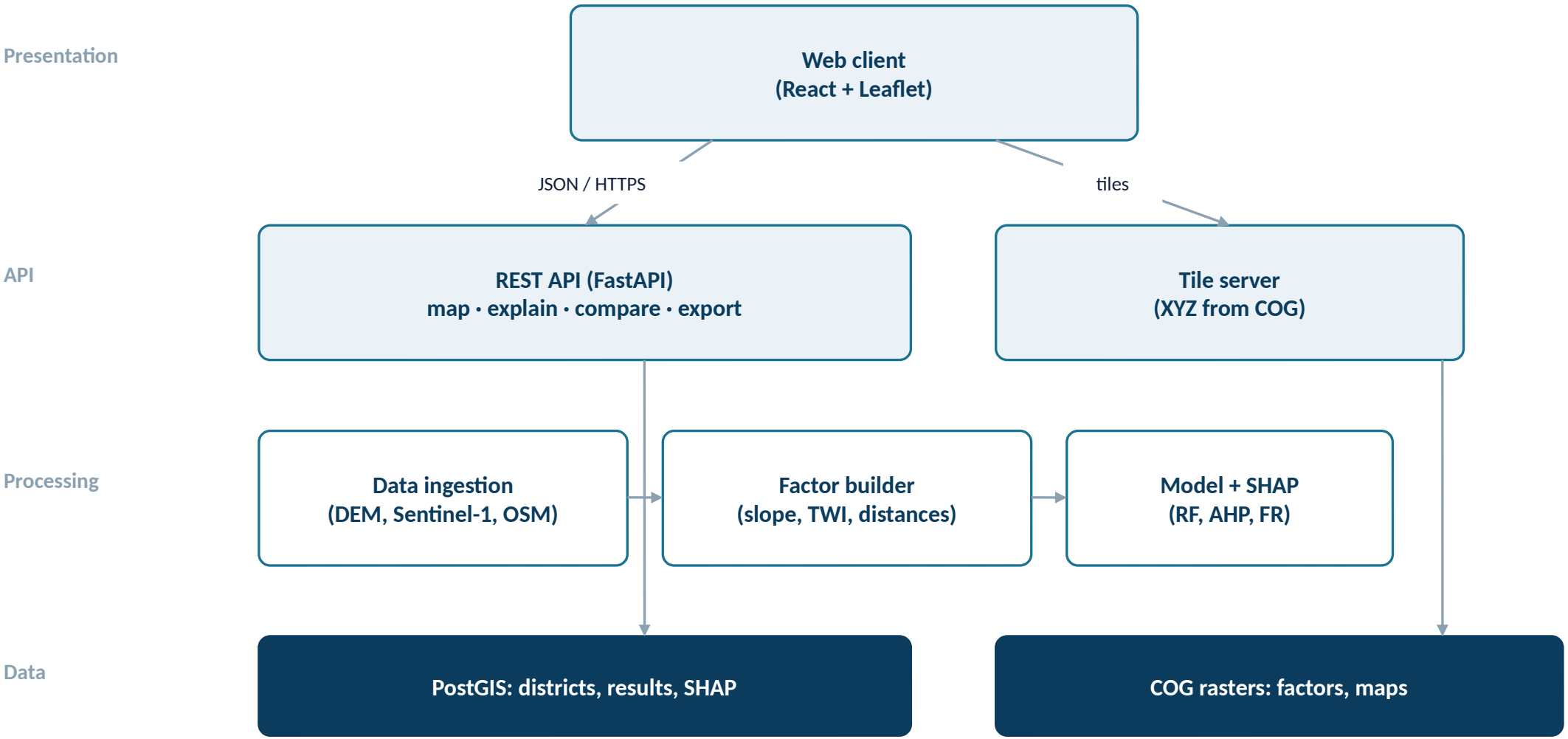
Researcher inherits Planner use cases.



Component diagram (first draft)

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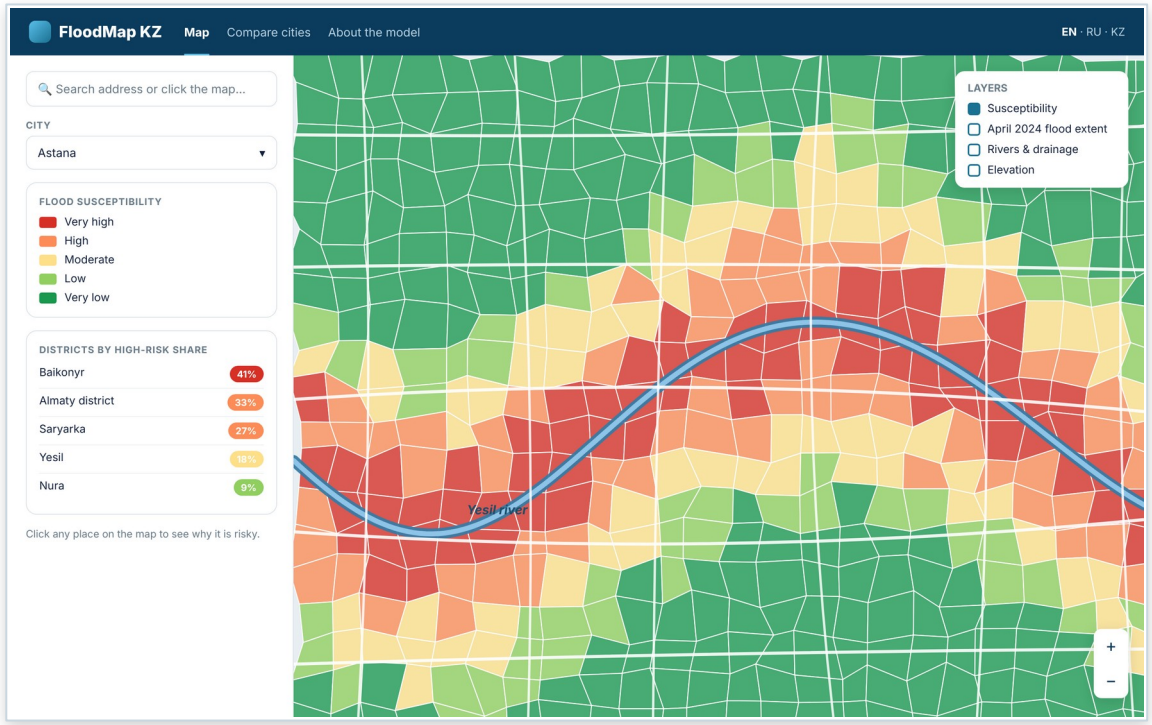


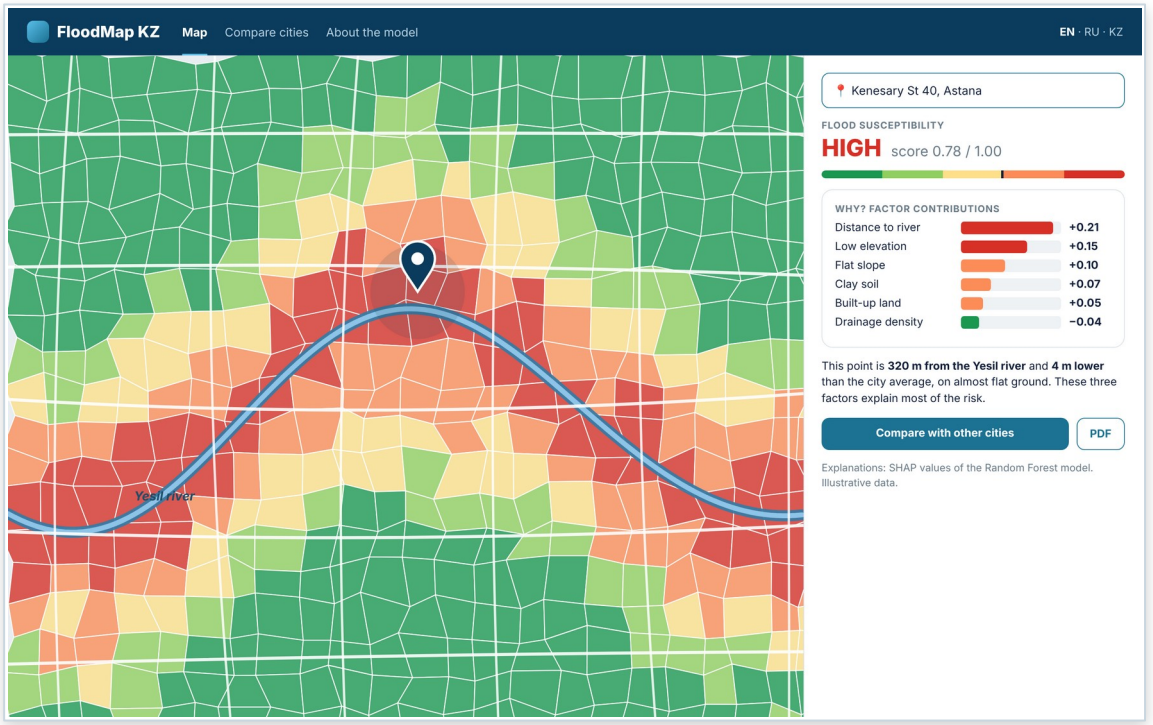
Processing runs offline as a Python pipeline and writes results to the data layer; the API only reads.

UX mockups: “Is my district at risk?”

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TASK





1

Open the city map

Districts are coloured by susceptibility class; the side list ranks districts by high-risk share.

2

Search or click a place

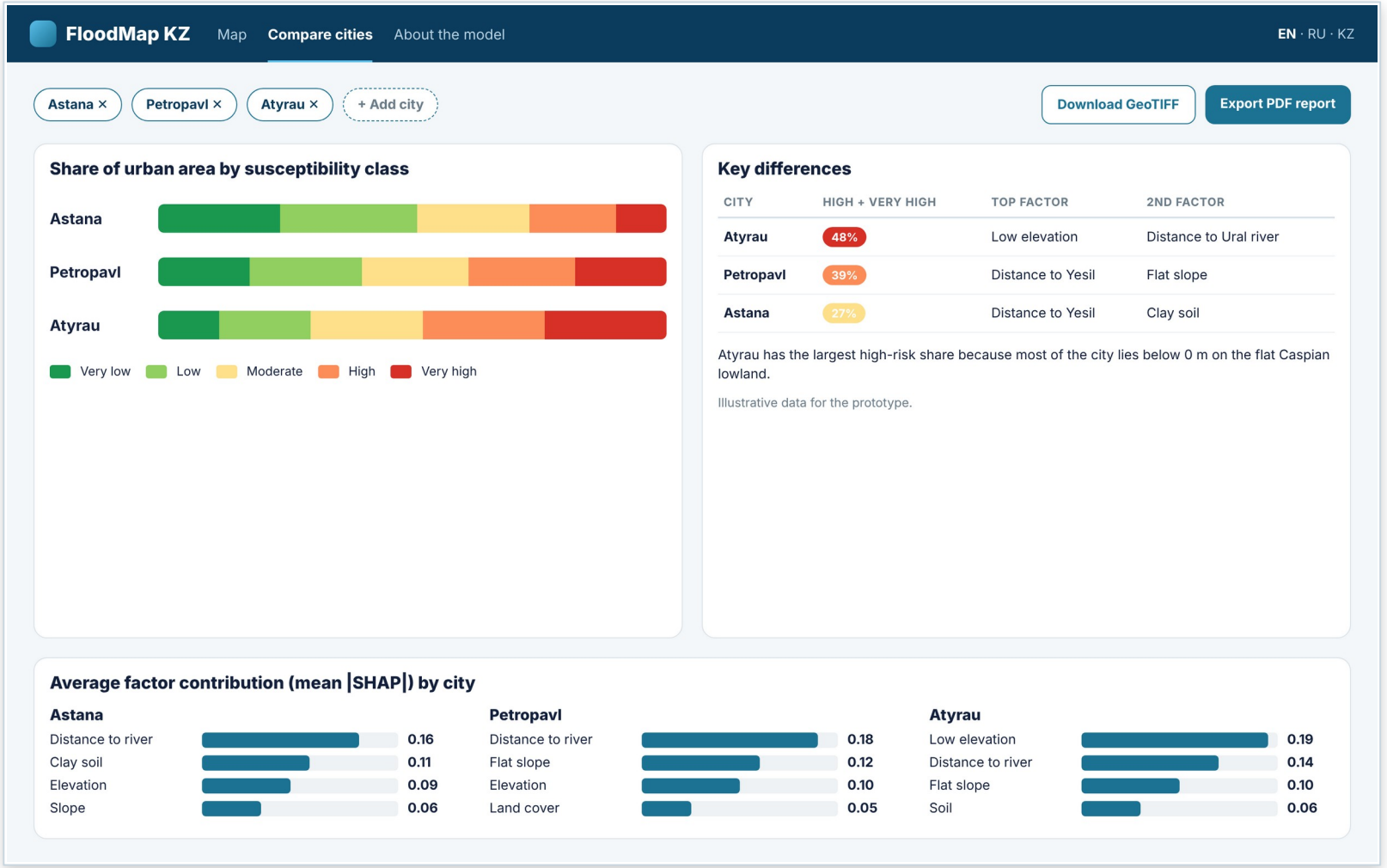
The panel shows the class, the score and the contribution of each factor (SHAP), plus a plain-language explanation.

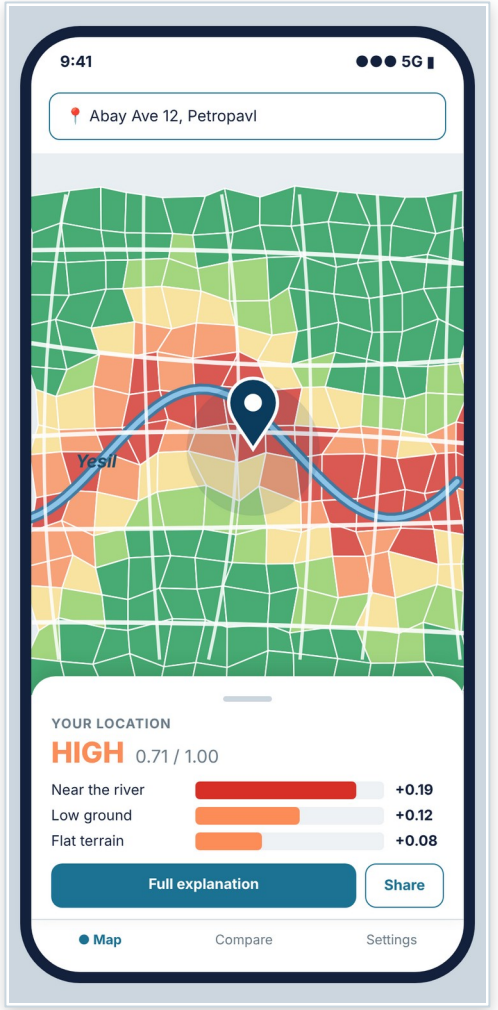
Target: a first-time user answers “is my district at risk and why” in under 60 s. Mockups use illustrative data.

UX mockups: compare cities and mobile view

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TASK





Mobile view: requested by survey respondents.

Scenario 2 (planner): add cities → compare high-risk share → see top factors → export PDF or GeoTIFF.

How we verify the model

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TASK

Aspect	Metric	Target	Measurement conditions
Discrimination	AUC-ROC	≥ 0.80	70/30 spatially blocked split of flood / non-flood points
Classification	Accuracy, F1, Kappa	F1 ≥ 0.75	Threshold at 0.5; balanced sample
Agreement with reality	Share of 2024 flooded area in High/Very high	≥ 70%	Sentinel-1 flood extents, April 2024
Explanation stability	Rank correlation of SHAP importances	≥ 0.8	10 bootstrap retrains
Sensitivity	Change of class map when AHP weights ±20%	≤ 15% cells	One-at-a-time weight perturbation

Baselines for comparison: (1) AHP-only model, (2) Frequency Ratio model, (3) global flood hazard maps (JRC / Aqueduct) resampled to districts. The main model must beat all three on AUC and on agreement with 2024 flood extents.

How we test the software

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TASK

Data pipeline	pytest unit tests for factor rasters: CRS, resolution, no-data, value ranges	**Coverage ≥ 70%**
REST API	RestAssured / pytest contract tests on every endpoint, OpenAPI schema check	**All endpoints green in CI**
Web UI	Playwright end-to-end tests of the 3 key scenarios in 3 browsers	**p95 tile load ≤ 1 s**
Usability	5+ users from the survey do 3 tasks, then fill SUS questionnaire	**SUS ≥ 70, task ≤ 60 s**

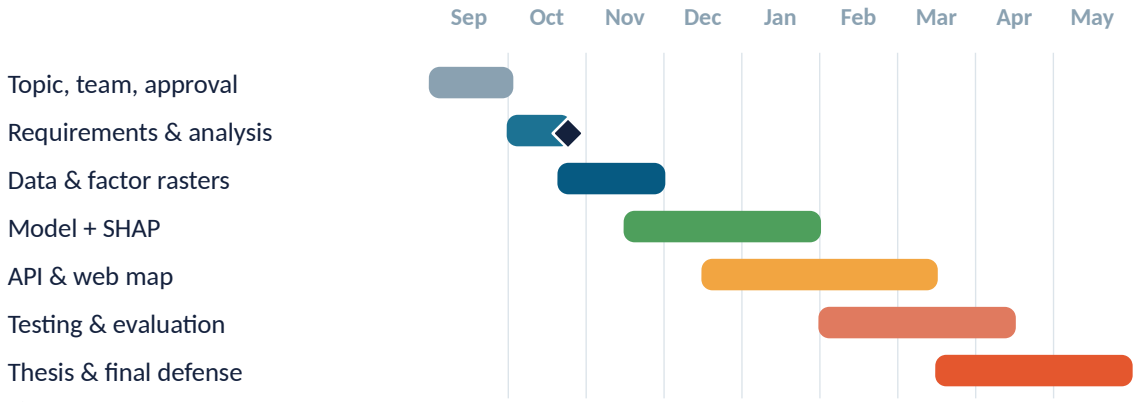
Tests run in GitHub Actions on every pull request; bugs tracked in Taiga with the task ID (TG-23).

Project Management and Risks

1 Team responsibilities

Member	Role	Main responsibilities
Rakhymbay Y.	QA engineer & developer	Data preparation, factor calculation (Python), testing, model validation, CI
Tenglesbayev Zh.	Developer	Model implementation, susceptibility maps, API and web map
Tutkabay A.	Analyst & PM	Literature, factor weighting, comparative analysis, schedule, supervisor contact

2 Project timeline (Sep 2026 – May 2027)



◆ *Pre-defense 1 (24 Oct 2026). Data before modelling; testing runs alongside development.*

3 GitHub & Taiga

GitHub

- One shared repo, personal accounts
- src/ docs/ diagrams/ tests/
- Branch per task, PR + review
- Tag pre-defense-1
- No secrets or personal data

github.com/W0ran/flood-susceptibility-kz

Taiga

- Board: To do / In progress / Review / Done
- Task IDs in branches and commits: TG-23-database-schema
- Sprints of 2 weeks
- 34 user stories in 5 epics

tree.taiga.io · Flood Susceptibility KZ

4 Project risks

Risk	Prob.	Impact	Mitigation
Too few flood points for training	Medium	High	Sentinel-1 2024 extents; AHP/FR baselines
Coarse data for KZ cities	Medium	High	Global open data; fix city list early
Overfitting (spatial autocorrelation)	Medium	Medium	Spatially blocked validation
Schedule slips (studies + work)	Medium	High	2-week sprints, weekly sync
Scope creep (real-time warnings)	High	Medium	MoSCoW; warnings = future work

Supervisor Feedback and Improvements

#	Supervisor's feedback (15–16 Oct)	Action taken	Status
1	**File is damaged, use case diagram does not open** PowerPoint asked to repair the file; send PDF	Found the cause: invalid shape settings on the use case slide and duplicate shape IDs. Fixed, every slide checked; PDF version attached	✓ Completed
2	**Full project title, remove slide 2** Title slide incomplete; agenda not needed	Official topic title on slide 1; agenda slide removed	✓ Completed
3	**No UX / prototype** Mockups of key user scenarios required	High-fidelity mockups of 4 screens and 2 scenarios (slides 17–18); clickable Figma prototype next	◔ In progress
4	**Literature review in one slide, full titles** Two slides with short references	One table: 12 publications with full titles, source, method and use	✓ Completed
5	**Add Project Management and Risks** Timeline Sep 2026 – May 2027	New slide: team roles, Gantt chart, GitHub & Taiga, risk register	✓ Completed
6	**No real GitHub and task tracker links** Slide 21 must link to working tools	Private repo and Taiga project (34 user stories, 5 epics); team and supervisor invited; links on slide 21	✓ Completed

Example of an implemented improvement: UX

Before: low-fidelity wireframes

Grey boxes, no real content or data



After: high-fidelity mockups

Real map, factor explanations, compare and mobile screens

